IngeTrazo Plugins Manual

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Introduction

Four plugins add building and printing tools to IngeTrazo: three make geometry, and one scales the result for a 3D printer.

Plugin	What it makes	Shortcut
Gutters & Downpipes	Full-size gutters, downpipes and fittings that join onto each other	Ctrl+Alt+G
3D Brick & Block Studio	3D brick or block faces on a wall face, with bonds, openings and soldier courses	Ctrl+Shift+T
Loft	A solid that blends between two or more shapes, such as a square into a circle	Ctrl+Alt+L
3D Print Scaler	A copy of your model at 1:50, 1:76 or any scale, checked against your print bed	Ctrl+Alt+P

A typical model goes through them in this order:

1. Draw the building at full size in IngeTrazo.
2. Add brickwork with **3D Brick & Block Studio**, rainwater goods with **Gutters & Downpipes**, and shaped parts such as chimney pots or transitions with **Loft**.
3. Make a scaled copy with **3D Print Scaler**.
4. Export the copy as STL and slice it.

All sizes you type are in millimetres. IngeTrazo stores geometry in metres internally, whatever your display units, and the plugins convert for you.

Installing and updating

Each plugin is one `.py` file, installed by copying it into IngeTrazo's plugins folder and restarting.

1. Open File Explorer, paste `C:\Users\eddy_\AppData\Roaming\ingetrazo\plugins` into the address bar and press Enter.
2. Copy the `.py` file into that folder. When updating, choose **Replace the file**.

3. Restart IngeTrazo. The plugins remember their settings only while IngeTrazo is open, so a restart also clears old settings.

File	Plugin
`gutters_downpipes.py`	Gutters & Downpipes
`brick_block_studio.py`	3D Brick & Block Studio
`loft.py`	Loft
`print_scaler.py`	3D Print Scaler

Keep only one copy of each plugin in the folder. Two copies under different names may both load.

Shortcuts: each plugin's shortcut is listed in the Introduction. Ctrl+Shift+P is not used because IngeTrazo already uses it for the Python Console.

Running from the Python console: Gutters & Downpipes, Loft and 3D Print Scaler can be started from the console. Paste the file's code in, select what the plugin needs, then type `run()`. The plugin finds the 3D view by itself. If it reports that it can't, use `run(viewport)`.

Gutters & Downpipes

Builds full-size, hollow or solid gutters, downpipes and fittings, each in its own group, and joins every new part onto the open end of the last. Press **Ctrl+Alt+G** to open its window. It stays open while you work, so you can add part after part.

The window

Area	What it does
Top line	Says where the next part will go, for example "joins onto the far end of: Gutter length"
Solid (for 3D printing)	Fills gutters and pipes in, with no hollow inside
Preset	Picks a saved set of sizes; **Save as**, **Delete** and **Open folder** manage your own presets
Part list	Gutter parts and downpipe parts
Gutter / Downpipe	Profile or shape, width or diameter, depth and wall thickness, in mm

Area	What it does
This part	A live 3D preview, plus the settings the chosen part uses
Start a gutter on a wall	Places the first gutter along the top of a wall you select
Brackets and clips	Automatic fascia brackets and pipe clips, and chunky fittings for printing
Buttons	**Add part**, **Use selected end**, **Start at the origin**, **Close**

In the preview, the brown panel is the wall and the pale grey part is the one the chosen fitting sits over.

Parts

Part	Settings it uses	Notes
Gutter length	Length	Gets fascia brackets automatically
Running outlet	Outlet length	Its spigot underneath takes a downpipe
Stop end	—	Closes a gutter run
Union bracket	Distance back from the end	Sits over the joint between two lengths
Fascia bracket	Distance back from the end	Its back plate sits on the wall
90° and 135° angles	Leg length	External corners turn towards the wall, internal corners away from it
Downpipe length	Length	Gets pipe clips automatically
Bend / offset bend	Turn angle, direction	Turn 67.5 = 112.5° offset bend, 87.5 = 92.5° bend, 45 = 135° bend
Branch (Y junction)	Turn angle, direction, branch length	The arm angles back up towards the incoming flow
Shoe	Direction	Kicks away from the wall by default; closes a downpipe run
Pipe socket / coupler	Distance back from the end	Sits over a pipe joint
Pipe clip	Distance back from the end	Its lug points at the wall

Gutter profiles are half-round, square-line, ogee and deep-flow. Downpipes are round or square.

Building a run

1. Choose a starting point. With nothing chosen, the first part goes at the origin; **New run - wall is** sets which side the wall is on. To start on a building, see **Starting on a wall** below.
2. Pick a part, check its sizes in the preview, and click **Add part**.
3. The next part joins onto the far end of the one you just added. Keep adding parts.
4. After a running outlet, choosing a downpipe part puts it on the outlet's spigot automatically.
5. A stop end or shoe finishes a run. The window then moves to the newest end still open, for example the outlet spigot, or the start of the gutter for the second stop end.

To join onto any other end, click that open end face in the model, then **Use selected end**. Unions, sockets and clips sit over a joint, and the next part still carries on from that joint.

You can move or rotate placed parts with IngeTrazo's own tools and still join onto them. If you undo a part, click an end and **Use selected end**, because the window still points at the removed part.

Starting on a wall

1. Select a wall face and click **Use selected wall**.
2. Pick a gutter part and click **Add part**. It runs along the top of the wall, starting from the left end as you look at the wall from outside.

Setting	Default	Effect
Below top of wall	25 mm	Gap from the top of the wall down to the top of the gutter
Extra gap from wall	0 mm	0 puts the brackets flat on the wall
Past left end / Past right end	0 mm	Positive runs past that end. Negative stops short of it, leaving room for an outlet, angle or stop end
Gutter length = wall length	On	The first gutter length fits the wall, including the two end settings
Wrong side?	Off	Tick it if the gutter appears on the inside of the wall

Brackets and clips

With **Add them evenly** ticked, each gutter length gets fascia brackets and each downpipe length gets pipe clips. One goes about 150 mm from each end, and the rest are spaced evenly between, no further apart than you set: 1000 mm for gutters and 1800 mm for pipes by default. A 5 m gutter gets 6 brackets; a 3 m downpipe gets 3 clips. They go in their own group, and a single undo removes them together with their length.

Chunky fittings (for 3D printing) makes every bracket, clip, union, socket and stop end thicker, 3× by default. It also closes the gap between each fitting and the part it fits over, and turns each fascia bracket into one solid block down to its wall plate. On a wall, the gutter moves out just enough for the chunkier brackets to stay flat against it.

Presets

Four presets are built in:

- UK half-round 112 + round 68
- UK square-line 114 + square 65
- Deep-flow 117 + round 68
- Cast iron ogee 115 + round 75

Save as stores your current sizes and options as one `.json` file in Documents › IngeTrazo Presets › Gutters and Downpipes. **Open folder** opens that folder, so you can copy or share preset files.

Limits

- The running outlet has no hole cut in the gutter floor; its spigot sits underneath.
- Joints are butt joints, not overlapping sockets.
- Joining works only on ends made by this plugin.

3D Brick & Block Studio

Covers a selected wall face in 3D bricks or blocks, cutting them neatly around window and door openings. Select the wall face and press **Ctrl+Shift+T**.

If you select several faces, the one with the most openings is used as the wall. The other selected faces mark which openings get a soldier course: select the face filling an opening along with the wall.

Sizes and presets

Every size box shows its real value in the wall's units (mm or m). The defaults are a 210 × 65 × 20 mm brick slip with 10 mm joints.

Built-in preset	Length × height × depth (mm)
Brick slip	210 × 65 × 20
UK standard brick	215 × 65 × 102.5
German NF brick	240 × 71 × 115
US modular brick	194 × 57 × 92
Concrete block	440 × 215 × 100
Concrete block	390 × 190 × 190

Save as stores your own sizes under a name; **Delete** removes one of your presets. Your presets are kept in the file `.brick_block_studio_presets.json` in your user folder.

Bonds and row offsets

- **Bond pattern:** running (half offset), stack, third, English or Flemish.
- **Row offsets:** follow the bond, use a fixed offset on every second course, repeat a custom sequence of fractions such as `0, 0.5, 0.25, 0.75`, or randomise each course. Random offsets use a seed, so the same seed gives the same wall, and they avoid joints lining up with the course below.
- **Header courses:** add a course of headers every Nth course, with its own header width.

Openings and soldier courses

Tick **Soldier course around openings** to stand bricks on end around each opening. The list shows every opening in the wall with its size and position; tick the ones that get a course, or use **All** and **None**. **Soldier length** follows the unit length until you change it. **Corner angle** sets how sharp a bend must be to count as a corner.

Other settings

Setting	Effect
Mortar joint	Gap between bricks; 0 means no joint
Min cut fragment	Cut pieces smaller than this are left out
Max subdivisions	Detail used when cutting around curved or concave openings
Include back faces	Closes the back of every brick, so each one is a closed solid
Swap width / height axes	Lays the courses the other way across the face
Put units in a new group	Keeps the bricks separate from the wall

For 3D printing, the bricks need a backing, otherwise the slicer fills the gaps across the joints with V shapes. Either tick **Include back faces**, or duplicate the wall face and group it with the bricks; both slice cleanly.

3D Print Scaler

Scales a selected object to a model scale and shows its printed size in millimetres before you commit. Select the object (a group, or all of its faces) and press **Ctrl+Alt+P**.

Setting	Effect
Scale	Presets from 1:1 to 1:1000, or **Custom…** for any scale such as 1:76.2. The custom box shows the chosen scale's number, so it starts from there
Model is drawn in	Leave this on Metres: IngeTrazo stores geometry in metres even when it displays millimetres. Marked **(guessed)** when the plugin guessed from the object's size; your own choice is remembered
Scale about	Bottom centre (stays on the print bed), centre, lowest corner, or the world origin
Mode	**Scale a copy** keeps the original and puts the copy beside it. **Scale in place** changes the original
Check against print bed	Your bed's X, Y and Z size in mm; the size line turns red if the copy won't fit, even turned 90°

The information line shows the current size and the printed size, for example "Printed size: 20.0 × 20.0 × 20.0 mm".

Scale in place refuses loose faces that share corners with geometry you didn't select, because moving them would distort the neighbours; group the object first or use **Scale a copy**. If any point can't be moved, the model is put back exactly as it was.

Detail size: at small scales, fine details shrink below what a printer nozzle can draw. A 10 mm mortar joint is 0.2 mm at 1:50, half the width of a standard 0.4 mm nozzle. Exaggerate small details before scaling, print larger, or use a finer nozzle.

Loft

Makes a closed, printable solid that blends between two or more flat shapes, such as a square at the bottom and a circle above it. Draw the shapes, select their faces, and press **Ctrl+Alt+L**. The solid goes in a new group; your original shapes are left in place to delete or hide.

The shapes must face each other, one above the other or one in front of the other. Two shapes lying side by side on the ground give an error. Only the outer edge of each shape is used; holes are ignored.

Setting	Effect
Points per profile	Points around each shape. Defaults to at least 64 for curved lofts; it can't go below the shape with the most corners
Sections between profiles	Bands up the solid: 1 for Straight, 32 for S-curve, 24 for Smooth by default
Path	**Smooth S-curve** (sides leave and meet each shape square-on), **Straight**, or **Smooth curve through all profiles** for 3 or more shapes
Line profiles up automatically	Matches corners to the nearest part of the next shape, so the solid doesn't twist
Extra twist	Turns the last shape by this many points, to add a twist on purpose
Cap the first / last profile	Closes the ends; both must be ticked for a printable solid
Side faces	Auto splits only clearly bent side faces into triangles
Smooth finish	Hides the edge lines between faces that meet at less than the set angle (30° by default)

Changing **Path** fills in that path's suggested numbers; you can type your own afterwards.

IngeTrazo shades each face flat, so a smooth surface can still look faintly faceted on screen. For a smoother look, raise the points to about 96 and the sections to about 48. The printed part follows the true shape either way.

Printing and troubleshooting

For small-scale prints, make every part a closed solid and make details large enough for your nozzle.

1. Build at full size.
2. In Gutters & Downpipes, tick **Solid** and **Chunky fittings**. In Brick & Block Studio, tick **Include back faces** or add a backing face.
3. Scale with **3D Print Scaler** using **Scale a copy**, and check the bed fit.
4. Export the copy as STL.

5. In the slicer, check the **sliced layer preview**, not just the 3D view, to see what will actually print.

Problem	Cause	Fix
A part comes out about 1000 times too big	Sizes treated as metres	Use the latest Gutters & Downpipes and restart IngeTrazo. In 3D Print Scaler, set Model is drawn in to Metres
V shapes or slivers in mortar joints in the slicer	Bricks open at the back	Tick **Include back faces**, or group a backing face with the bricks
Mortar joints vanish when printed	Joints thinner than the nozzle	Widen joints before scaling (15–30 mm), print larger, or use a 0.2–0.25 mm nozzle
The plugin's shortcut is missing	Another command uses it, or the file isn't in the plugins folder	Check the shortcut in Preferences › Keyboard shortcuts, put the file in the plugins folder, and restart
A downloaded .py file won't open in Windows	Windows has no program set for .py, or blocked it	Right-click › Open with › Notepad, or Properties › Unblock; to install, just copy it into the plugins folder
A new part doesn't join where expected	The window still points at an undone or deleted part	Click the right open end and use **Use selected end**
The gutter appears inside the wall	The wall face points inwards	Tick **Wrong side?** in Start a gutter on a wall
"Could not mark the open ends" in the console	IngeTrazo didn't keep the joining marker on the faces	That part can't be joined onto; add the next part from the origin and move it into place